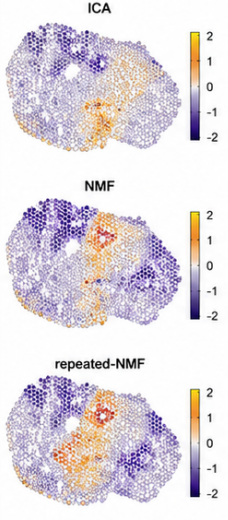
ORIGIN-Ax: origin-constrained graph refinement of latent axes

Expression similarity is refined by origin compatibility and conflict penalties before family-level claim decisions.

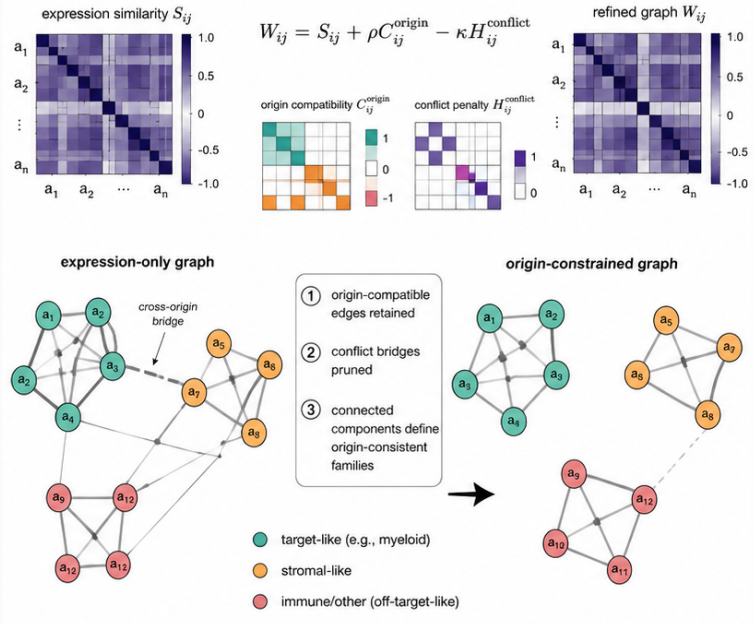
A. *Candidate latent axes are noisy and mixed-origin*



same biology may appear as redundant or mixed-origin candidate axes

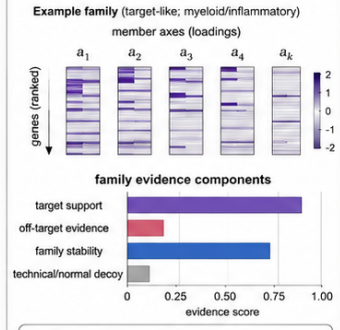
single axis a_j = (loading l_j, score z_j)

B. *Origin-constrained graph refinement*



graph ablation example: *families 2 -> 3; target truth hits 1 -> 2*

C. *Family evidence and identifiability*



$$M(F) = E_{\text{target}} - E_{\text{normal}} - E_{\text{off}}$$

$$\Delta(F) = E_{\text{target}} - \max(E_{\text{normal}}, E_{\text{off}}, E_{\text{technical}})$$

held-out genes and optional anchors are audit evidence, not primary scoring.

D. *Claim-controlled family outcomes*

target-supported — passes identifiability margin and off-target control

off-target — coherent biology but better explained by a competing origin

non-identifiable — insufficient or conflicting evidence

